# Analytics & Big Data

Session 5: Regression II

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🎯 **Learning objectives**

- **Explain** how business problems can be formulated as binary classification tasks and modeled using logistic regression.
- **Understand** how logistic regression uses a linear predictor and sigmoid function to produce probabilities, apply thresholds, and interpret log-odds and coefficients.
- **Evaluate** classification models using confusion matrices and metrics such as accuracy, precision, recall, and F1 score.
- **Apply** model predictions to decision-making by selecting appropriate thresholds based on business costs and expected value.

# Business problem: Churn prediction

# Predicting customer churn

**Why churn prediction matters**

A lost customer is often a **predictable** customer.

Firms collect many traces of customer relationships — usage, spending, support interactions, contract data.
The analytical challenge is to turn these traces into an **early warning system**.

**Typical variables in a churn dataset**

- **Who is the customer?** tenure, segment, contract type, subscription tier
- **How do they use the service?** login frequency, feature usage, inactivity days
- **How satisfied are they?** complaints, support tickets, response times, NPS/satisfaction
- **Are there warning signs?** payment delays, downgrades, cancellations of add-ons, price sensitivity

**Prediction goal**

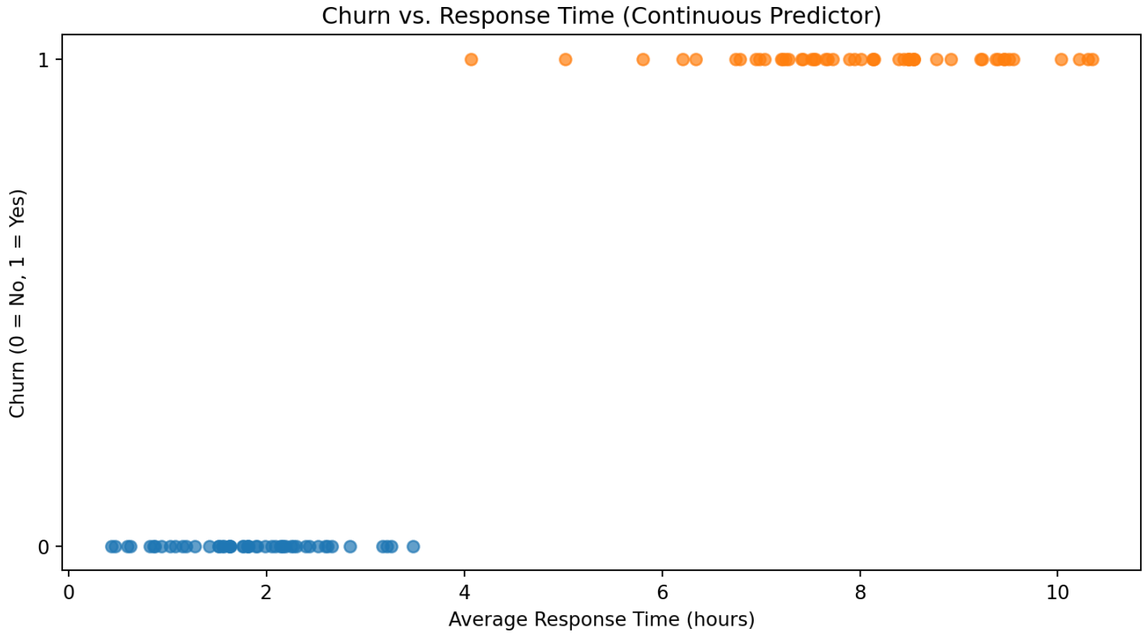
Predict:

$$P(\text{churn} = 1 \mid X)$$

where $X$ contains the observed customer characteristics and behaviors.

# Geometric view of the problem

Churn prediction is a **binary classification problem**:




Possible approach: We could use Linear Regression with a threshold at `0.5` to classify output:

- If $f(x) < 0.5$, predict `y=0`
- If $f(x) \geq 0.5$, predict `y=1`

# Logistic regression

# Classification

Examples:

- Online transactions: Fraudulent (yes/no)?
- Customer churn: yes/no?
- EMail: Spam/not spam?

$$y \in \{0, 1\}$$

with

0: “Negative class” (e.g., no fraud, no churn, no spam)
1: “Positive class” (e.g., fraud, churn, spam)

# Logistic regression model

Ideally, our model should predict the **probability** of `y=1`. This would allow us to apply a threshold for classification, but it would also give us more information (how likely an observation belongs to the *positive class*).

This means we need a model that ensures continuous values between 0 and 1:

$$y \in [0, 1]$$

**Step 1: Linear model**

$$z_i = \beta_0 + \beta_1 x_i$$

- Same structure as linear regression
- $z_i \in \mathbb{R}$ (can be any real number)

**Step 2: Transform to probability via sigmoid function**

$$\mathbb{P}(y_i = 1 \mid x_i) = \sigma(z_i)$$

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

**Logistic regression model**

$$\mathbb{P}(y_i = 1 \mid x_i) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_i)}}$$

# Log-odds

Logistic regression assumes that the **log-odds of the outcome are linear in the predictors**:

$$\log\left(\frac{y_i}{1 - y_i}\right) = \beta_0 + \beta_1 x_i$$

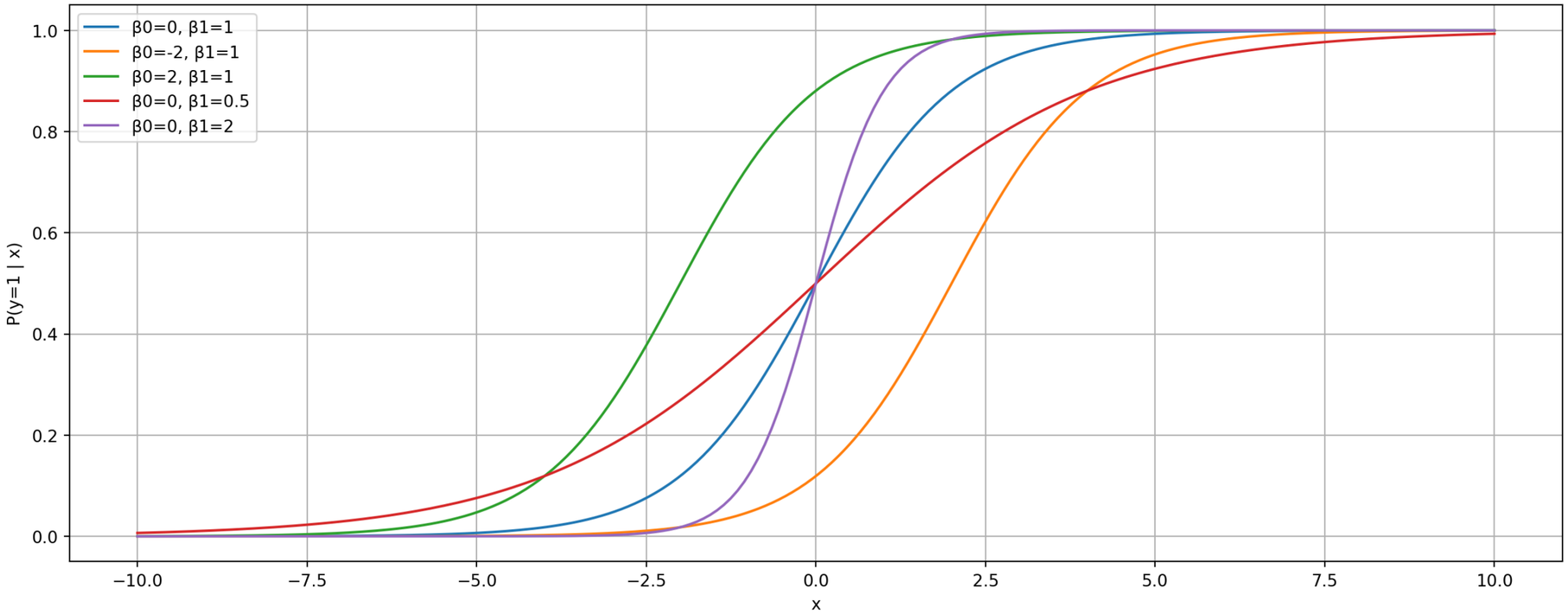
*Note:* $\frac{y_i}{1-y_i}$ *are the "odds".*

This implies a **sigmoid-shaped relationship** between x and the y:

$$y_i = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_i)}}$$

The sigmoid function serves as a *link function* to convert the linear predictor into a *probability*.

# The Sigmoid function




**Note**

- Changing $\beta_0$ **shifts** the curve left/right
- Changing $\beta_1$ **changes slope** (flatter vs. steeper transition)

# Estimation: Maximum likelihood

Instead of minimizing squared errors (like in linear regression), logistic regression chooses parameters that make the **observed data most likely**.

**Core idea**

We want predicted probabilities to match observed outcomes:

- If $y_i = 1$ → predicted probability should be **high**
- If $y_i = 0$ → predicted probability should be **low**

**Likelihood function**

For each observation:

$$P(y_i \mid x_i) = p_i^{y_i}(1 - p_i)^{1-y_i}$$

For all observations:

$$L(\beta) = \prod_{i=1}^{n} p_i^{y_i}(1 - p_i)^{1-y_i}$$

- **Maximization**: Choose $\beta_0, \beta_1$ to maximize this likelihood

**Intuition:** We search for the model that assigns **high probability to what actually happened**.

# Interpretation of coefficients

Logistic regression coefficients are **not directly effects on probability**.

They describe effects on the **log-odds**.

**Coefficient meaning**

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 x$$

- $\beta_1$ : change in **log-odds** when $x$ increases by 1

**More intuitive: odds ratios**

Exponentiate the coefficient:

$$e^{\beta_1}$$

- Factor change in the **odds**
- Example:
  - $e^{\beta_1} = 1.5$ → odds increase by **50%**
  - $e^{\beta_1} = 0.7$ → odds decrease by **30%**

**Important implication**

- Effect on probability is **nonlinear**

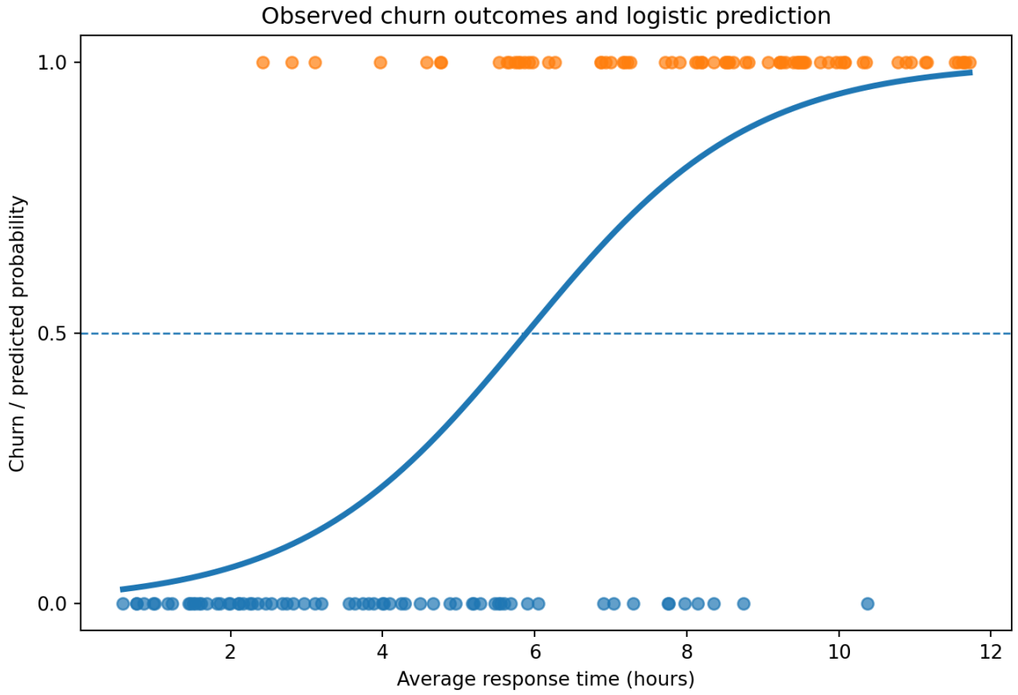
# Evaluation

# Predicted probabilities

A logistic regression model predicts a **probability** of churn:

$$P(\text{churn} = 1 \mid x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}$$

> A threshold (such as **0.5**) is needed to convert probabilities into class labels.
>
> Comparing predicted vs. actual class labels allows us to evaluate the performance of a classifier like logistic regression.



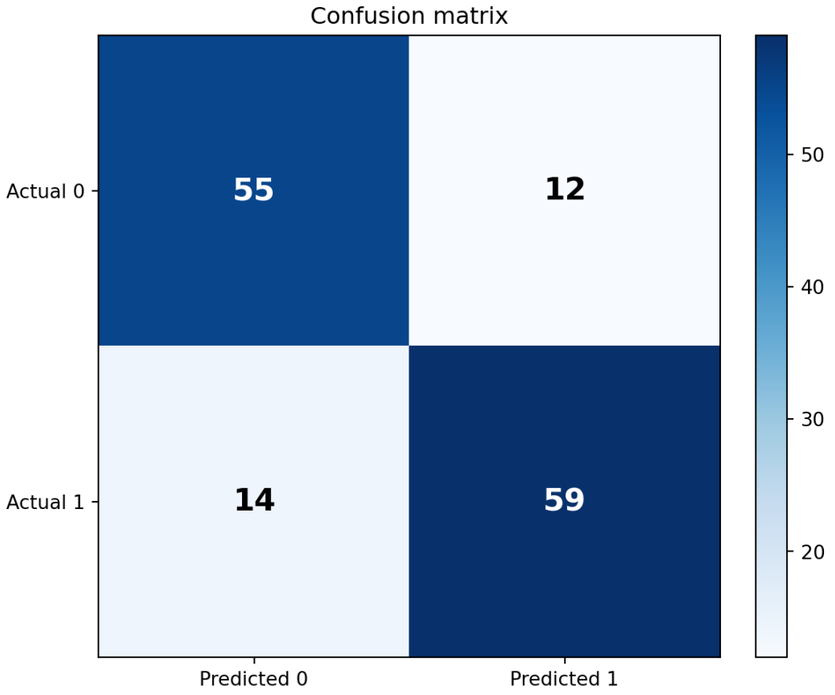

Note: there are no meaningful *residuals* (or $R^2$), as in linear regression models.

# Confusion matrix

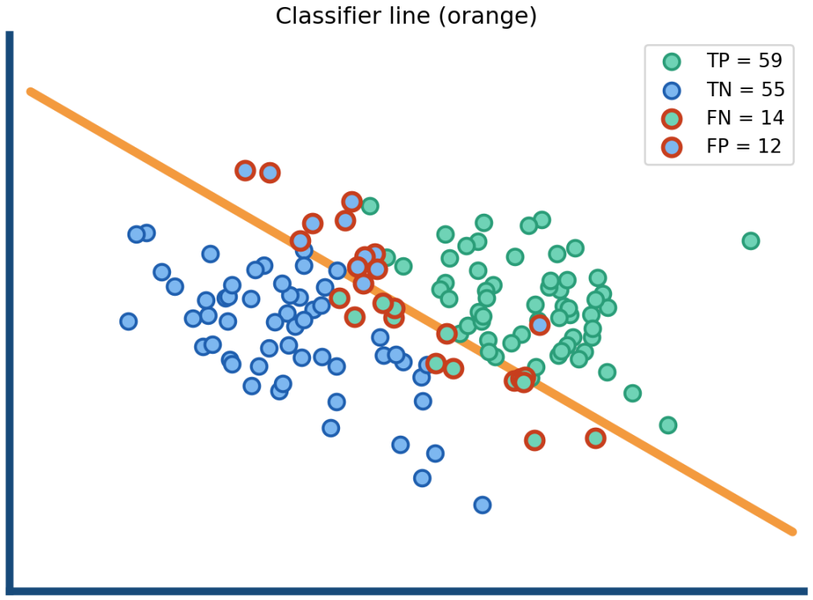
Using a **0.5 threshold**, predicted probabilities are turned into classes.

The confusion matrix summarizes:

- **True negatives (TN):** predicted no churn, observed no churn
- **False positives (FP):** predicted churn, observed no churn
- **False negatives (FN):** predicted no churn, observed churn
- **True positives (TP):** predicted churn, observed churn



# Evaluation metrics




| Metric | Formula | Interpretation |
|---|---|---|
| **Accuracy** | $\frac{TP+TN}{TP+TN+FP+FN}$ | Overall share of correct predictions |
| **Precision** | $\frac{TP}{TP+FP}$ | Among predicted positives, how many are actually positive? |
| **Recall** | $\frac{TP}{TP+FN}$ | Among actual positives, how many did the model identify? |
| **F1** | $2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision}+\text{Recall}}$ | Balance between precision and recall |

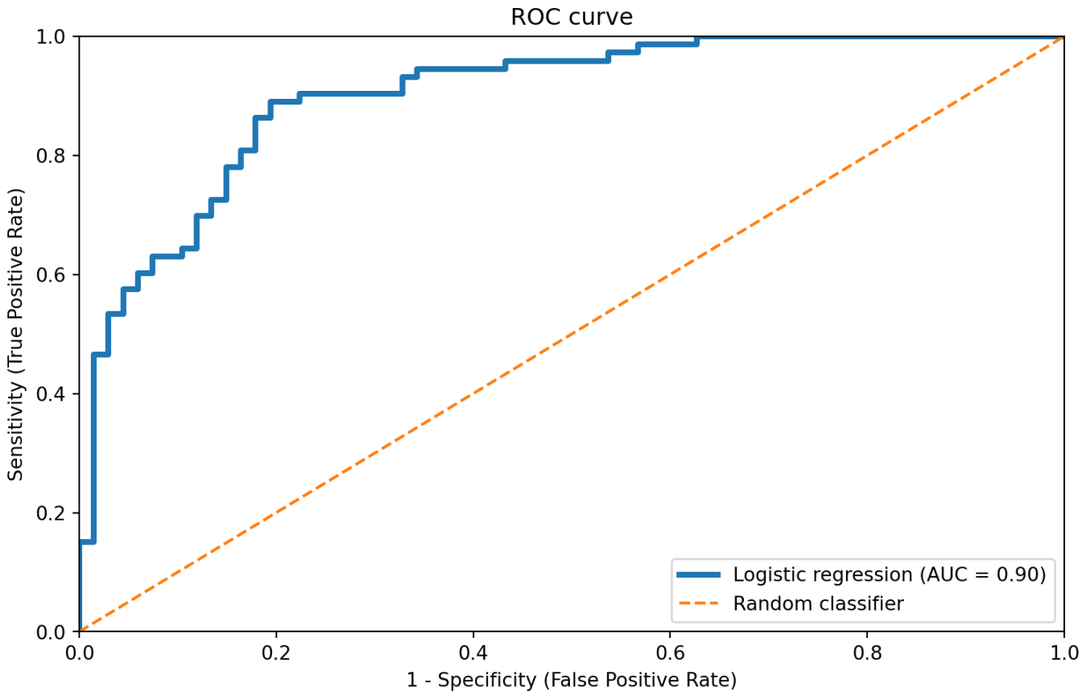
# ROC curve and AUC

**ROC** (receiver operating characteristic) curve summarizes the trade-off between:

- **Sensitivity / Recall / True Positive Rate**
- **False Positive Rate**

As we vary the **classification threshold**, the model becomes more or less conservative in predicting `churn = 1`.

> **Area under the curve (AUC)** summarizes ROC performance in a single number:
>
> - **AUC = 0.5**: no better than random guessing
> - **AUC = 1.0**: perfect separation
> - **Higher AUC**: better ranking of churners vs. non-churners




A good classifier achieves:

- **high true positive rate**

Unlike accuracy, ROC/AUC evaluates the classifier across **all possible thresholds**.

- **low false positive rate**

# From prediction to decisions

# From prediction to decision

A logistic regression model predicts a **probability of churn**:

$$p(x) = P(\text{churn} = 1 \mid x)$$

This probability is based on **historical data**:

- It reflects how similar customers behaved in the past
- It captures **baseline churn risk**
- It is **not a certainty**

Suppose a firm has two possible actions:

- **Intervene**: offer a retention incentive
- **Do nothing**

So the business problem is no longer only:

> Who is likely to churn?

It becomes:

> For whom is an intervention worth it?

# Decision-making under uncertainty

We can use the expected value criterion to **translate predictions into decisions** by combining:

- **Predicted churn probability** from the model
- **Business consequences** of the action

A simple payoff structure:

| Action | Customer would churn | Customer would stay |
|---|---:|---:|
| Intervene | +20 | -10 |
| Do nothing | -100 | 0 |

Interpretation:

- Saving a likely churner creates value
- Intervening with a customer who would stay creates unnecessary cost
- Missing a true churner is very costly

# Expected value of (not) intervening

| Action | Customer would churn | Customer would stay |
|---|---|---|
| Intervene | +20 | -10 |
| Do nothing | -100 | 0 |

If we **intervene**, the expected value is:

$$EV(intervene) = p(x) \cdot 20 + (1 - p(x)) \cdot (-10) = 30p(x) - 10$$

Interpretation:

- With probability $p(x)$, the customer is at risk and intervention helps
- With probability $1 - p(x)$, the customer would stay anyway and intervention wastes resources

If we **do nothing**, the expected value is:

$$EV(do_nothing) = p(x) \cdot (-100) + (1 - p(x)) \cdot 0 = -100p(x)$$

Interpretation:

- If the customer would churn, the firm bears a large loss
- If the customer would stay, there is no cost

Based on Elkan (2001).

# Optimal decision rule

Instead of selecting one action, we should intervene whenever:

$$EV(intervene) > EV(do_nothing)$$

Insert the expected values:

$$p(x) \cdot 20 + (1 - p(x)) \cdot (-10) > -100p(x)$$

Simplify:

$$20p(x) - 10 + 10p(x) > -100p(x)$$

$$130p(x) > 10$$

$$p(x) > 0.077$$

# Decision threshold

This gives a clear rule:

> **Intervene if**
>
> $$(x) > 0.077$$
>
> That is: intervene if predicted churn probability exceeds **7.7%**.

The threshold is low because:

- Losing a customer is **very costly**
- Intervening is **relatively cheap**

So even a small churn probability can justify action.

💡 A classification threshold does not have to be 0.5. It should depend on the **business context**.

# General decision rule

More generally, the threshold can be written as:

$$t = \frac{C_{01}}{C_{01} + C_{10}}$$

where:

- $C_{01}$ = cost of unnecessary intervention
- $C_{10}$ = cost of missing a churner

Decision rule:

$$\text{Intervene if } p(x) > t$$

# Two sources of knowledge

This decision combines two different inputs:

**1. Data-driven component**

- Logistic regression estimates:

$$p(x) = P(\text{churn} = 1 \mid x)$$

- Learned from historical data

**2. Business-driven component**

- The threshold

$$t = \frac{C_{01}}{C_{01} + C_{10}}$$

- Based on business costs, strategy, and domain knowledge

# Big conceptual takeaway

$$\textbf{Data} \rightarrow \textbf{Model} \rightarrow \textbf{Probability} \quad + \quad \textbf{Business costs} \rightarrow \textbf{Threshold} \rightarrow \textbf{Decision}$$

The model does **not** decide what to do.

It produces probabilities.

The **business defines the threshold**, and together they determine the action.

# Why probabilities matter

Probabilities create a clean interface between **prediction** and **decision-making**.

With probabilities:

- the output is interpretable
- the threshold can come directly from business costs
- model and decision rule remain conceptually separate

With arbitrary scores:

- interpretation is less clear
- thresholds must often be tuned empirically
- the connection to business value is less direct

# Outlook: Machine learning

# From statistical models to machine learning

Logistic regression is already a **classification model**:

- Predicts binary outcomes (0/1)
- Produces **probabilities** (P(y=1 X))
- Uses a **threshold** to assign class labels

It is:

- **Interpretable**
- **Fast and robust**
- Often a **baseline model**

**Logistic regression is often the first model to try.**

# From simple models to flexible models

| Logistic regression | Machine learning models |
|---|---|
| Linear decision boundary | Can learn nonlinear boundaries |
| Strong assumptions | Fewer assumptions |
| Few variables (selected) | Many variables (high-dimensional) |
| Interpretable coefficients | Often less interpretable |

Key shift:

- From **structured, theory-driven models**
- To **flexible, data-driven models**

# New challenges in machine learning

**1. Overfitting**

- Models may fit noise instead of signal
- Good training performance ≠ good generalization

**2. Role of data**

- More observations and features become important

**3. Evaluation**

- Same metrics as before:
  - Accuracy, precision, recall, F1
- But evaluated on **new/unseen data**

**4. Model tuning**

- Not only the classification threshold
- Additional **hyperparameters** must be chosen

# Summary

- Logistic regression is a method for **binary classification**, where the goal is to predict whether an outcome occurs (e.g., churn vs. no churn).
- It predicts **probabilities**

$$P(y = 1 \mid X)$$

  using a linear predictor combined with a **sigmoid function**, ensuring outputs between 0 and 1.
- Coefficients are interpreted in terms of **log-odds** (or odds ratios), not direct probability changes.
- Parameters are estimated via **maximum likelihood**, aiming to match predicted probabilities with observed outcomes.
- Model performance is evaluated using **classification metrics** such as confusion matrices, accuracy, precision, recall, F1, and ROC/AUC.
- A **threshold** is required to convert probabilities into class labels, and should be chosen based on **business costs**, not fixed at 0.5.

# Survey: Session 5

https://forms.gle/mVWw3z7ftFn48gDY6

Note: Responses may be analyzed and published in anonymized form.

Please complete the survey before you leave today — thank you 🙏

# References

Elkan, C. (2001). The foundations of cost-sensitive learning. *International Joint Conference on Artificial Intelligence*, *17*, 973–978. https://doi.org/10.5555/1642194.1642224